



2026-09-25 21:26 · 3 samples · 3 ok, 0 with warnings, 0 failed · operator: Example

Sample	Spoligotype	Octal	Species	Lineage	Pattern (spacers 1 to 43)	Status
AF2122_97	SB0140 SIT683	664073777777600	M. bovis	BOV		OK
NC_002945	SB0140 SIT683	664073777777600	M. bovis	BOV		OK
H37Rv	Spoligo not found SIT451	777777477760771	M. tuberculosis	4.9		OK

Reviewed by		Date		Signature	
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Species and lineage

Region	Deleted in	Depth relative to MTBC control	Call
RD1	BCG, Dassie bacillus	0.80	present
RD4	M. bovis, BCG (some M. canettii)	0.00	deleted
RD7	M. africanum lineage 6, animal lineages	0.00	deleted
RD9	M. africanum (lineages 5, 6), animal lineages	0.00	deleted
RD12	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)	0.00	deleted

Reads per spacer

Blue: present (count ≥ 5). Orange: called absent but seen in some reads.

Species and lineage

Region	Deleted in	Depth relative to MTBC control	Call
RD1	BCG, Dassie bacillus	1.00	present
RD4	M. bovis, BCG (some M. canettii)	0.00	deleted
RD7	M. africanum lineage 6, animal lineages	0.00	deleted
RD9	M. africanum (lineages 5, 6), animal lineages	0.00	deleted
RD12	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)	0.00	deleted

Contigs per spacer

Blue: present (count ≥ 1). Orange: called absent but seen in some contigs.

Species and lineage

Region	Deleted in	Depth relative to MTBC control	Call
RD1	BCG, Dassie bacillus	1.00	present
RD4	M. bovis, BCG (some M. canettii)	1.00	present
RD7	M. africanum lineage 6, animal lineages	1.00	present
RD9	M. africanum (lineages 5, 6), animal lineages	1.00	present
RD12	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)	1.00	present

Contigs per spacer

Blue: present (count ≥ 1). Orange: called absent but seen in some contigs.

Run information

Operator	Example
User	example
Computer	example
Operating system	Linux-6.8.0-139-generic-x86_64-with-glibc2.39
Started	2026-09-25 21:26:38 EDT
Finished	2026-09-25 21:26:50 EDT
Duration	12.1 s
Working directory	/example
Command	spoligotyper -i data/ -o results/ -t 8 --operator Example

Parameters

Input	/example/data
Output folder	/example/results
Minimum count	default (5 for fastq, 1 for fasta)
Threads	8
Jobs	1
Seal memory	1g
Species and lineage	yes

Software

spoligotyper	0.5.0
Python	3.14.7 (/opt/conda/bin/python3.14)
Seal	/opt/conda/bin/seal.sh
BBTools	40.02
Java	openjdk version "25.0.2-internal" 2026-01-20

Reference data

Spoligotype database	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligotype_db.txt 1,976 patterns · MD5 0a88f7b89f746de2a11e1b4af0151e36
Spacer sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligo_spacers.fasta 43 spacers · MD5 033c08e66aaa90193973f8d542e94c1
Species markers	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/markers.fasta MD5 87135cd99f26936911997867c6f1964b
Lineage SNP barcode	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_barcode.tsv MD5 f676fb8f470889ae02cb2f2b6c5970d7
Lineage SNP sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_snps.fasta MD5 7e662a9baee450e79065705eb7b79021
SIT database	/opt/spoligotyper/sit_database.tsv 9,656 patterns, 3,850 SITs · SHA-256 ba32dfccad2f1167580754779a42fd8ea480c444060c1f2144a51b1b2b43e3ac SIT database for spoligotyper, converted from SpolLineages Spoligo_list.csv (SITVIT2 patterns)

Method

Spacers were counted with Seal (BBTools): each of the 43 spacers is searched as a single 25-mer on both strands, allowing 1 mismatch ($k=25$, $hdist=1$, $rcomp=t$, $maskmiddle=f$, $ambiguous=all$). A spacer is present when it is found in at least the minimum count of reads (contigs for assemblies). The binary pattern is converted to the octal code (Dale et al. 2001) and the hexadecimal code, and looked up in the spoligotype database for its SB number. Species: the read depth of regions of difference RD1, RD4, RD7, RD9 and RD12 (100 bp segments, same Seal parameters) is compared with the depth of MTBC-specific control regions; a region is deleted when its relative depth is at most 0.1, present when it is at least 0.5, and the species is read from the RD profile as in the RD PCR scheme, refined with the lineage SNPs and the spacers. The fraction of MTBC reads is the control depth divided by the depth expected from the number of bases. Lineage: reads carrying each allele of the 62 SNPs of the Coll et al. (2014) barcode are counted with exact 31-mers ($k=31$, $hdist=0$); a lineage is called when at least 80% of the reads (and at least 3, or 1 contig) carry its allele.

References

Kamerbeek J et al. Simultaneous detection and strain differentiation of *Mycobacterium tuberculosis* for diagnosis and epidemiology. *J Clin Microbiol* 35:907-914 (1997). doi:10.1128/jcm.35.4.907-914.1997

Dale JW et al. Spacer oligonucleotide typing of bacteria of the *Mycobacterium tuberculosis* complex: recommendations for standardised nomenclature. *Int J Tuberc Lung Dis* 5:216-219 (2001).

Smith NH, Upton P. Naming spoligotype patterns for the RD9-deleted lineage of the *Mycobacterium tuberculosis* complex; www.Mbovis.org. *Infect Genet Evol* 12:873-876 (2012). doi:10.1016/j.meegid.2011.08.002

Brosch R et al. A new evolutionary scenario for the *Mycobacterium tuberculosis* complex. *Proc Natl Acad Sci USA* 99:3684-3689 (2002). doi:10.1073/pnas.052548299

Couvin D, Segretier W, Stattner E, Rastogi N. Novel methods included in SpolLineages tool for fast and precise prediction of *Mycobacterium tuberculosis* complex spoligotype families. *Database (Oxford)* 2020:baaa108. doi:10.1093/database/baaa108 (SIT database, from SITVIT2)

Coll F et al. A robust SNP barcode for typing *Mycobacterium tuberculosis* complex strains. *Nat Commun* 5:4812 (2014). doi:10.1038/ncomms5812

Bushnell B. BBTools. <https://sourceforge.net/projects/bbmap/>

Duceppe M-O. spoligotyper 0.5.0: in silico spoligotyping of *Mycobacterium tuberculosis* complex genomes. Zenodo. <https://doi.org/10.5281/zenodo.22926160>